

Superset Determinations in Modal Triplet Theory: Parameter Identifiability after Profile-Standard SM Closure

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Version 3

Abstract

We formulate the MTT superset program as a parameter-identifiability framework. Embedded renormalized-Standard-Model equivalence closes at the adopted one-shared-physical-primitive and measured-profile standard using selected SMDR v1.3 transport and a complete eight-row covariance. The construction ledger contains one shared electroweak primitive and 15 measured SM profile coordinates; transported couplings, nine charged Yukawa rows, the Higgs row, covariance entries, and the 96-dimensional finite Dirac operator are outputs and are not counted again as independent inputs. Exact source-side structure includes the native faithful gauge group, anomaly-free chiral carrier, completed finite triple, one-Higgs projector, and 10/10 finite gauge/ghost spectrum rows. None of these presently replaces the measured profile by a zero-primitive source theorem. We classify every result as a structural theorem, calibration, profile/replay, diagnostic, or held-out prediction and state the resulting identifiability limits.

Version 3 Revision Note

Supersedes *Superset Determinations in Modal Triplet Theory*, version 2.

Reason One-loop crossings, fitted zeta ratios and threshold vectors, and the claimed $\alpha_s(M_Z)$ cross-prediction reused target-related data and were not independent determinations. The earlier parameter discussion also risked double-counting transported outputs as independent inputs.

Resolution Version 3 replaces the numerical claim chain by a parameter-identifiability ledger tied to the selected SMDR profile and classifies every quantity by theorem, calibration, replay, diagnostic, or prediction status. It incorporates the native finite-geometry results and separates finite-replay Yukawa closure from strict source-value emission.

Retained result The superset strategy remains useful for comparing encodings, exposing shared primitives, and discovering common-source constraints.

Remaining boundary Predictive promotion requires a selected continuum source/action that replaces the measured profile and supports genuinely held-out observables.

1 Purpose of the superset program

MTT can organize several effective encodings through shared carrier, projector, bundle, overlap, and response data. The superset strategy is useful when it asks which observables are jointly representable and which parameters are identifiable from a declared input set. It is not, by itself, a method for deriving measured constants.

We use five disjoint status classes:

1. **Structural theorem:** follows from selected mathematical data without observed value selection.
2. **Calibration:** parameters are chosen to reproduce declared data.
3. **Profile/replay:** measured coordinates are transported through a specified scheme or embedded action.
4. **Diagnostic:** a comparison tests consistency but was involved in selection or fitting.
5. **Held-out prediction:** the compared observable was absent from source selection, calibration, branch choice, and uncertainty construction.

2 Retirement of the legacy Tier-3 execution

The former scales Λ_{12} and Λ_{23} were obtained from a one-loop crossing exercise. The reported values near 4.2–5 TeV were then used to extract ζ ratios, fit a common scale, choose a minimum-norm threshold, and reconstruct α_s . This chain is retired.

Specifically:

- a gauge crossing is not an MTT coherence or cutoff scale;
- the old 0.560 and 0.229 ratios are not current profile values;
- K was calibrated with gravitational/electroweak information and was not an independent prediction;
- the minimum-norm threshold was a regularized fit selected by an external norm, not a source theorem; and
- α_s was not held out from the latent extraction and threshold choices, so it was not a cross-prediction.

3 Current common-scheme profile

The selected numerical transport uses SMDR v1.3 at

$$Q = M_t = 172.5590883453979 \text{ GeV}.$$

Its gauge entries are

$$\begin{aligned} g_Y &= 0.3585945042 \pm 0.0000307251, \\ g_2 &= 0.6475986708 \pm 0.0000287665, \\ g_3 &= 1.163427409 \pm 0.004036156. \end{aligned}$$

Fifteen locked source coordinates are transported into eight full-SM $\overline{\text{MS}}$ rows. The differentiated map emits a positive-definite 8×8 covariance with all 36 symmetric entries and all 15 BCT–WZH cross-block entries. These rows are profile transport, not MTT-derived constants.

4 Adopted closure and parameter count

The final global audit closes 12/12 obligations for embedded renormalized-SM equivalence at the adopted standard. The independent-input ledger is:

Class	Count	Interpretation
Construction-side continuous primitive	1	shared P_{EW} , counted once
Measured SM profile coordinates	15	empirical common-scheme source profile with declared covariance policy
Neutral extension coordinates	2	measured oscillation splittings, outside the 12/12 SM baseline
Higgs-specific construction parameters	0	no separately fitted Higgs primitive

The eight SMDR output rows, nine charged Yukawa magnitude rows, finite 96×96 Dirac matrix, threshold/mass-scheme rows, and propagated covariance are transported or reconstructed outputs. They are not additional independent inputs and must not be counted a second time. Conversely, calling them outputs does not make the 15 measured source coordinates predictions. The current ledger therefore makes no strict parameter-reduction claim relative to the Standard Model.

The distinction between two closure standards matters here. The nine charged Yukawa rows close at the accepted finite-replay/profile standard, and the Higgs lane closes with zero Higgs-specific parameter once P_{EW} is declared. At the stronger zero-primitive tier, however, the observed magnitudes and the value of P_{EW} are not emitted by selected source geometry. Thus strict source replacement remains open without reopening the profile-standard Yukawa or Higgs result.

5 Selected structural outputs

The current superset contains genuine source-side structure independent of the profile values, including:

- the locked 27-by-27 qutrit–Weyl/minimal matrix architecture;
- the selected rank-two literal Cech–HYM witness, closed 2/2;
- the internal $K_{\text{threshold}}$ response ledger, closed 10/10;
- the selected retarded $q79/F/m1$ orientation representative; and
- the CKM II rows, closed at the prediction-with-uncertainty-profile standard.
- the native faithful $(SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$ gauge group and anomaly-free 48-state chiral carrier;
- the completed 96-dimensional finite triple at profile tier and its single anomaly-free shared hypercharge circle;
- the exact selected one-Higgs projector; and
- all ten finite gauge/ghost spectrum-source rows, with zero continuous parameters.

These achievements constrain how the measured profile can be embedded. They do not automatically emit every profile magnitude. In particular, the normalized finite gauge spectra are universal and therefore cannot determine the observed relative gauge couplings without a selected continuum response.

6 Strict-upgrade ledger

Beyond the adopted baseline, nine stronger obligations are tracked:

ID	Upgrade	Status
U1	zero-primitive empirical source	open
U2	literal global Cech–HYM/QaSU3	closed 2/2
U3	official joint input likelihood	partial
U4	CKM prediction beyond source rows	closed at profile criterion
U5	absolute neutrino mass and ontology	partial
U6	strong-CP selection	partial
U7	MTT-derived quantization	partial
U8	constructive nonperturbative 4D QFT	partial
U9	unique observed-branch selection	partial

Closing U2 or U4 does not imply that the other seven upgrades close.

7 Identifiability theorem

Theorem 7.1 (Current-ledger identifiability). *Given the selected branch, the 15-coordinate common-scheme profile, one shared electroweak primitive, the embedded SM action, and the selected finite operator data, the accepted observable rows are jointly reproducible with their declared covariance. If the measured profile is removed while the current source maps are held fixed, the physical numerical rows are not identifiable: the exact finite data do not supply a replacement map to the missing dimensionful and relative-coupling coordinates.*

Proof. The selected transport and embedding provide the forward map and covariance at the profile standard. The finite-replay Yukawa and Higgs constructions consume that profile and the declared shared primitive; the native finite gauge spectra are universal after lane normalization. No accepted current map sends the selected continuum geometry and upper action to all 15 measured source coordinates. Therefore the profile is sufficient for the adopted forward execution and presently necessary for its physical numerical values. The statement is about the current source inventory, not an impossibility theorem for future MTT constructions. $\square$

8 Interpretation of the shared structure

The superset does achieve something more specific than a relabeling of Standard Model parameters. One discrete carrier architecture supports gauge, fermion, Higgs, threshold, and flavor constructions with common projectors and provenance. This creates cross-sector consistency tests: a proposed source cannot be changed independently in one sector if the same object acts in the others. The native gauge quotient, anomaly table, finite triple, and one-Higgs projector are examples of such shared restrictions.

What the superset has not yet achieved is a generative numerical law for the measured profile. A compact representation with shared objects may be structurally economical without using fewer empirical coordinates at the strict prediction tier. The appropriate comparison with the Standard Model is therefore two-column: derived structural constraints on one side, and empirical continuous inputs on the other. Combining those columns into a single “parameter count” would obscure rather than demonstrate progress.

9 Rules for future superset calculations

Every future execution must publish raw inputs, units, scheme and scale, selection rules, fitted parameters, covariance, branch choices, code, and unit tests. A claimed held-out prediction must

include a data-separation certificate showing that the observable was absent from all source and calibration paths. No matching point, internal spectral gap, or gauge crossing may be promoted to a physical cutoff without a separate source theorem.

10 Conclusion

The superset program survives, but in a sharper form. Its current success is the coherent, reproducible embedding of the renormalized SM and the reduction of many sector-specific constructions to shared finite operator data. The native gauge, chiral, finite-triple, one-Higgs, and finite-spectrum results add real source-side restrictions without changing the empirical-input count. The old crossing-scale calibration and its downstream numerical claims are retired.

The next frontier is not another backfit. It is source promotion: emit the ten magnitude rows, relative gauge metrics, electroweak normalization, and physical threshold/mass values from one selected continuum action, or retain their measured source coordinates transparently. That distinction is the correct measure of progress toward strict no-knob closure.

Computational Evidence and Reproducibility

The listed packets are used directly as the finite family of profile, exact, and certified-numerical examples on which the identifiability analysis operates. Their values are not rederived here, and a profile row remains profile replay. The open ledger states the unresolved reduction from those effective coordinates to a strict no-knob source.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper's local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A01/direct_k_higgs_row (*derived exact*).
Promoted direct K_threshold.Omega_H.lambda row.
- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.

- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.
- A01/strict_pew_row (*derived exact*).
Promoted P_EW source row at the declared one-shared-primitive standard.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

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